Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– Ho Chi Minh City, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
2026-10-20	M	Nguyễn Trung Nam	Updated Content Status lifecycle to include archived status.
2026-10-20	M	Nguyễn Trung Nam	Replaced Business Rules BR-01 to BR-12.
2026-10-20	M	Nguyễn Trung Nam	Updated NFR section for roles and Vietnamese interface language.
2026-10-20	M	Nguyễn Trung Nam	Updated database entity summary to match smgs_schema.dbml.
2026-10-20	M	Nguyễn Trung Nam	Updated Application Messages MSG25-MSG29 and MSG35 to Vietnamese.
2026-10-20	M	Nguyễn Trung Nam	Updated cover page location and date to Ho Chi Minh City, October 2026.

*A - Added M - Modified D - Deleted

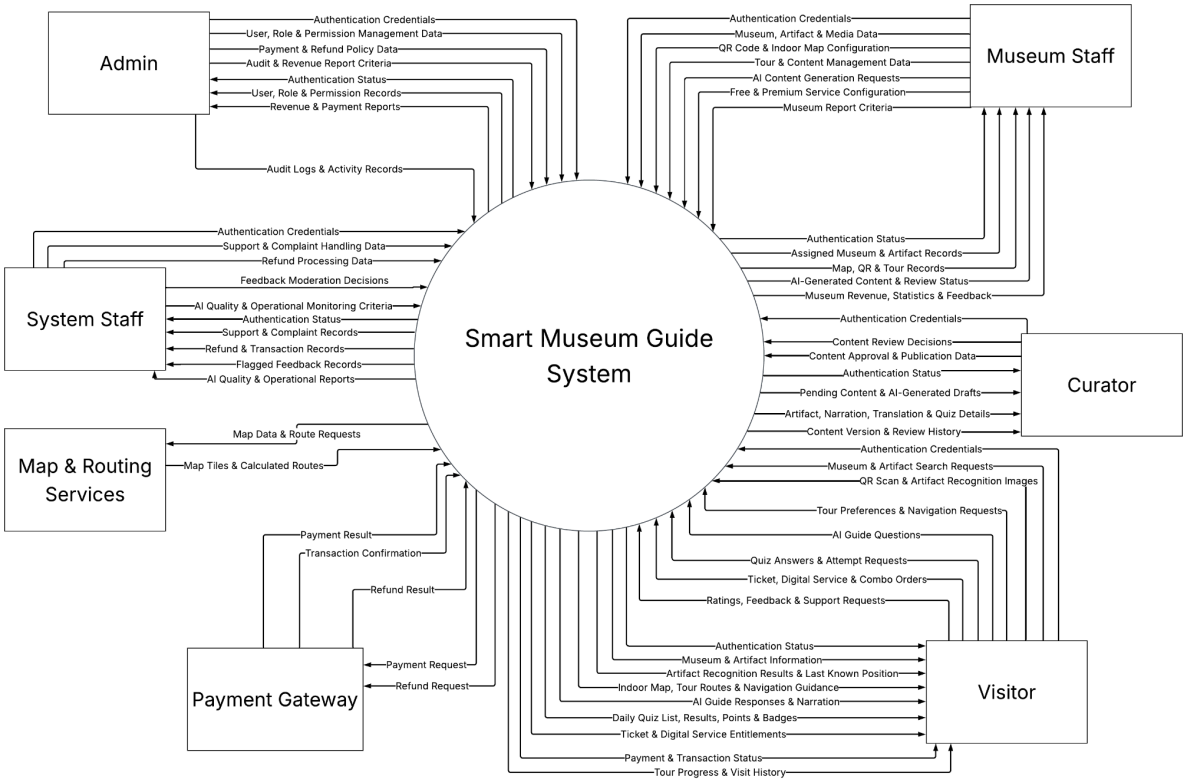
II. Software Requirement Specification

1. Product Overview

Smart Museum Guide System (SMGS) is a digital museum guidance platform consisting of a Flutter mobile application for Visitors and a role-based web portal for operational users. Visitors use the mobile application to discover supported museums, identify artifacts by QR code or image recognition, access published artifact information, listen to AI-assisted narration, ask contextual questions, take optional quizzes, receive personalized tours, obtain outdoor directions through Google Maps, purchase supported products, and submit feedback or support requests. The shared web portal is used by Administrators, Museum Staff, Curators, and System Staff. Museum Staff manage museum-scoped operational data (museum info, artifacts, QR codes, indoor maps, tours, feedback, and revenue) and create content drafts. Curators review, reject, or publish submitted museum content drafts. System Staff handle visitor support, complaints, refunds, transaction reconciliation, feedback moderation, operational incidents, and AI-quality monitoring. Administrators manage accounts, roles, staff assignments, supported museums, public Google Maps visibility, system settings, and audit records. Content status follows the exact lifecycle: draft, pending_review, published, rejected, and archived. The current release provides Vietnamese user interfaces by default, while multilingual public interface switching is reserved for future scope.

Out of Scope:

- A separate public web application for Visitors.
- Public multilingual user-interface switching in the current release.
- Automatic approval or automatic publication of AI-generated content.
- Hardware ticket scanners, turnstiles, and physical sensor-network integration.
- Redemption of points or badges for monetary rewards or physical goods.
- Storage of raw bank-card information or payment credentials.



2. User Requirements

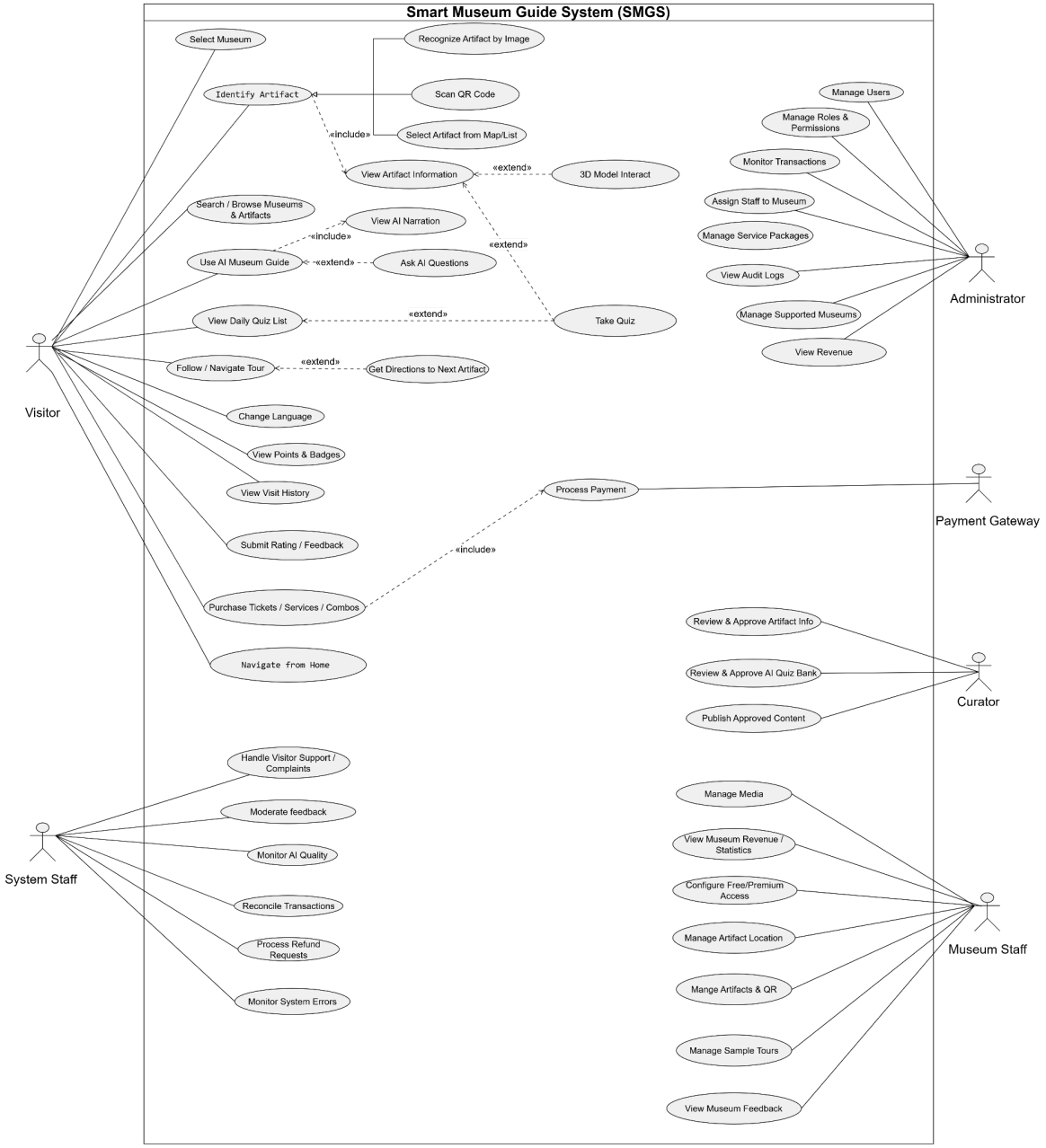
2.1 Actors

#	Actor	Description
1	Administrator	The Administrator manages user accounts, roles, staff assignments, supported museums, public museum visibility for Google Maps, system configuration, and audit records.
2	Visitor	The Visitor uses the SMGS mobile application to discover museums and artifacts, access published museum content, use AI guidance, take optional quizzes, receive personalized tours, purchase supported products, view personal history, submit feedback, and create support or complaint requests.
3	Museum Staff	The Museum Staff manages data within assigned museums, including museum information, artifacts, QR codes, indoor maps, tours, feedback, and revenue. Museum Staff can create AI-assisted content drafts and submit them for Curator review.
4	Curator	The Curator reviews museum-scoped artifact content, narration, and quiz submissions. The Curator can record review notes, reject unsuitable

		content drafts, and publish approved content within assigned museum scope.
5	System Staff	The System Staff handles visitor support, complaints, refund requests, transaction reconciliation, feedback moderation, operational incidents, and AI-quality monitoring across the system platform.
6	AI Service	The AI Service is an external service that produces draft narration, quiz suggestions, artifact-recognition results, and contextual answers. AI-generated museum content must be reviewed before publication.
7	Payment Gateway	The Payment Gateway is an external system that processes payment requests and returns payment results for supported ticket, digital-guide, and combo products.
8	Google Maps	Google Maps is an external service used to provide outdoor directions to museums that are publicly enabled for Visitor directions.

2.2 Use Cases

2.2.1 Diagram(s)



2.2.2 Descriptions

ID	Use Case	Actors	Use Case Description
UC-01	Select Museum	Visitor	Allows the Visitor to select a museum before accessing museum-specific artifacts, exhibitions, services, and other functions.
UC-02	Identify Artifact	Visitor	Allows the Visitor to identify an artifact using available identification methods such as image recognition,

			QR code scanning, or selecting an artifact from the map or list.
UC-03	Search / Browse Museums, Exhibitions & Artifacts	Visitor	Allows the Visitor to search and browse available museums, exhibitions, and artifacts.
UC-04	View Artifact Information	Visitor	Allows the Visitor to view published information about an identified or selected artifact, including free content and purchased premium content.
UC-05	Use AI Museum Guide	Visitor	Allows the Visitor to interact with the AI Museum Guide to obtain information and guidance about museum artifacts.
UC-06	Take Quiz	Visitor	Allows the Visitor to optionally complete quizzes related to museum artifacts and receive learning results at their own pace, including after exploring exhibits or after the museum visit. The quiz is supplementary and is not required during the physical tour.
UC-07	Get Personalized Tour	Visitor	Allows the Visitor to obtain a personalized museum tour based on their selected requirements and interests.
UC-08	Navigate Inside Museum	Visitor	Allows the Visitor to follow the museum's available navigation information while visiting the museum.
UC-09	Get Outdoor Directions	Visitor	Allows the Visitor to open Google Maps directions to active museums publicly enabled for directions.
UC-10	View Visit History	Visitor	Allows the Visitor to review records of previous museum visits and related activities.
UC-11	View Quiz Results	Visitor	Allows the Visitor to view the results of completed quizzes.
UC-12	Submit Rating / Feedback	Visitor	Allows the Visitor to submit ratings and feedback about museum visits or system services.
UC-13	Purchase Premium Digital Service	Visitor	Allows the Visitor to purchase premium digital services provided through SMGS.
UC-14	Manage Users	Administrator	Allows the Administrator to manage user accounts within the SMGS platform.
UC-15	Manage Roles & Permissions	Administrator	Allows the Administrator to configure system roles and access permissions.

UC-16	Monitor Transactions	Administrator	Allows the Administrator to monitor payment and transaction activities performed through SMGS.
UC-17	Manage Premium Digital Services	Administrator	Allows the Administrator to manage premium digital services available on the SMGS platform.
UC-18	Assign Staff to Museum	Administrator	Allows the Administrator to assign staff members to the appropriate museum.
UC-19	View Audit Logs	Administrator	Allows the Administrator to view system activity records for monitoring and auditing purposes.
UC-20	Manage Supported Museums	Administrator	Allows the Administrator to manage museums supported by the SMGS platform, including configuring the googleMapsVisible flag for Google Maps outdoor directions.
UC-21	View Revenue	Administrator	Allows the Administrator to view revenue information generated from SMGS services and transactions.
UC-22	Manage Basic Artifact Records	Museum Staff	Allows Museum Staff to create and maintain basic artifact records and translations for the assigned museum.
UC-23	Manage Ticket Type	Museum Staff	Allows Museum Staff to create, edit, and configure ticket types and their settings.
UC-24	Verify / Check-in Ticket	Museum Staff	Allows Museum Staff to verify visitor tickets and record museum check-in.
UC-25	Manage Artifact Information	Museum Staff	Allows Museum Staff to create draft artifact records, update draft artifact details, and manage media within assigned museum scope.
UC-26	Manage Exhibition / Gallery	Museum Staff	Allows Museum Staff to manage exhibition and gallery information for assigned museum scope.
UC-27	Manage Media	Museum Staff	Allows Museum Staff to upload and attach media files to artifact drafts within assigned museum scope.
UC-28	Manage Artifact Location	Museum Staff	Allows Museum Staff to manage artifact indoor-map positions and QR codes within assigned museum scope.
UC-29	Manage Narration Script	Museum Staff	Allows Museum Staff to draft narration scripts for museum artifacts before Curator review.
UC-30	View Visitor Feedback	Museum Staff	Allows Museum Staff to view ratings and feedback submitted for their assigned museum.
UC-31	Review / Edit AI Content	Curator	Allows the Curator to review, edit, reject, or record review notes on submitted artifact content and AI drafts within assigned museum scope.

UC-32	Approve / Reject Content	Curator	Allows the Curator to reject submitted content drafts with review notes or set content status to published within assigned museum scope.
UC-33	Publish AI Content	Curator	Allows the Curator to publish content with status pending_review, making artifact information, narration, and quizzes available to Visitors within assigned museum scope.
UC-34	Generate AI Narration	Museum Staff	Allows Museum Staff to request AI-assisted draft narration for an artifact before Curator review.
UC-35	Generate AI Quiz Content	Museum Staff	Allows Museum Staff to request AI-assisted quiz suggestions for an artifact before Curator review.
UC-36	Handle Support & Complaints	System Staff	Allows System Staff to inspect visitor support requests, respond to inquiries, and resolve complaints.
UC-37	Process Refund Requests	System Staff	Allows System Staff to review refund applications, verify transaction validity, and execute or decline refund operations.
UC-38	Reconcile Transactions	System Staff	Allows System Staff to reconcile local payment records against payment gateway settlement statements and correct discrepancies.
UC-39	Moderate Feedback	System Staff	Allows System Staff to review reported visitor feedback, hide inappropriate content, and maintain public feedback quality.
UC-40	Monitor Operations & AI Quality	System Staff	Allows System Staff to track system operational logs, monitor AI response quality, and log platform incidents.

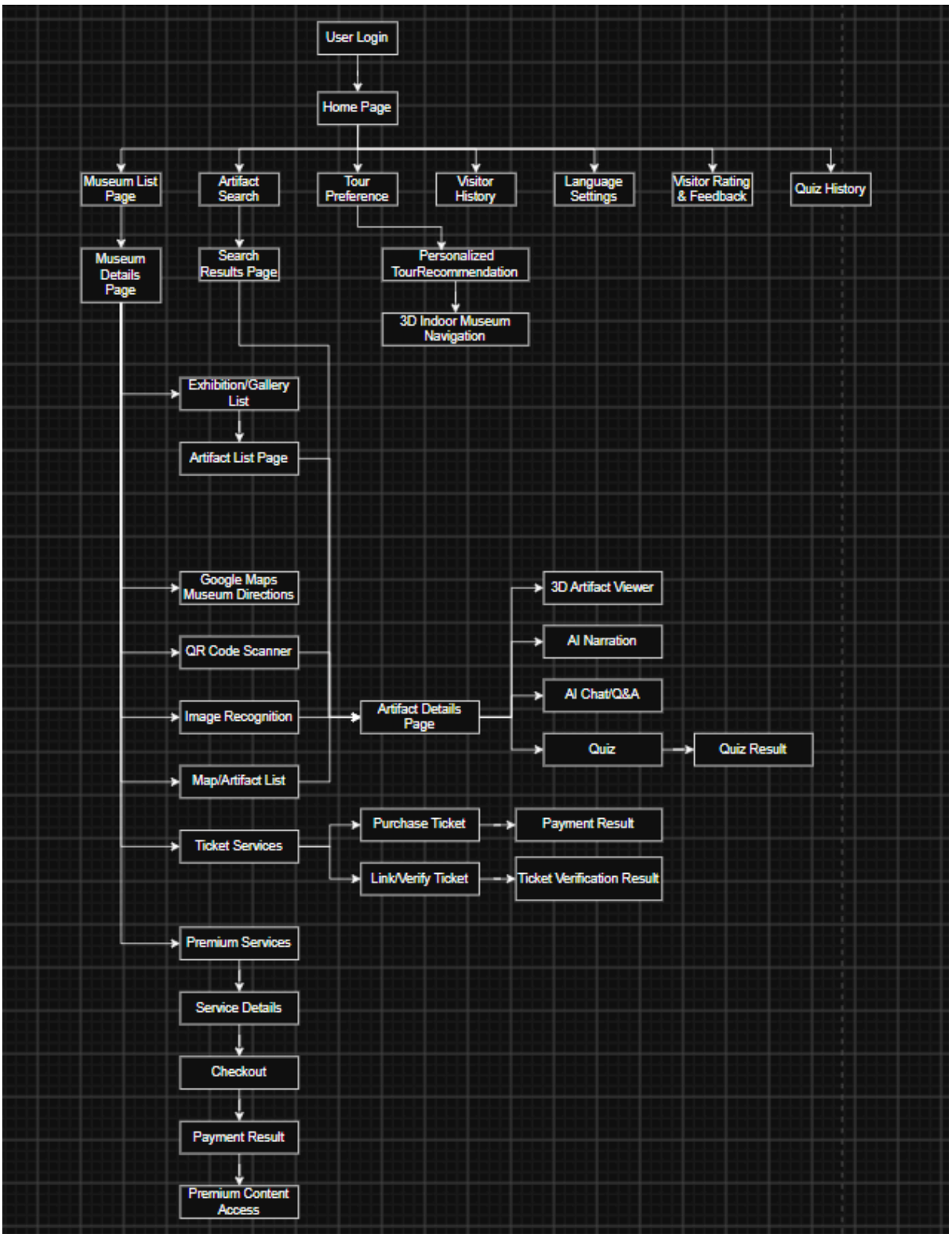
3. Functional Requirements

3.1 System Functional Overview

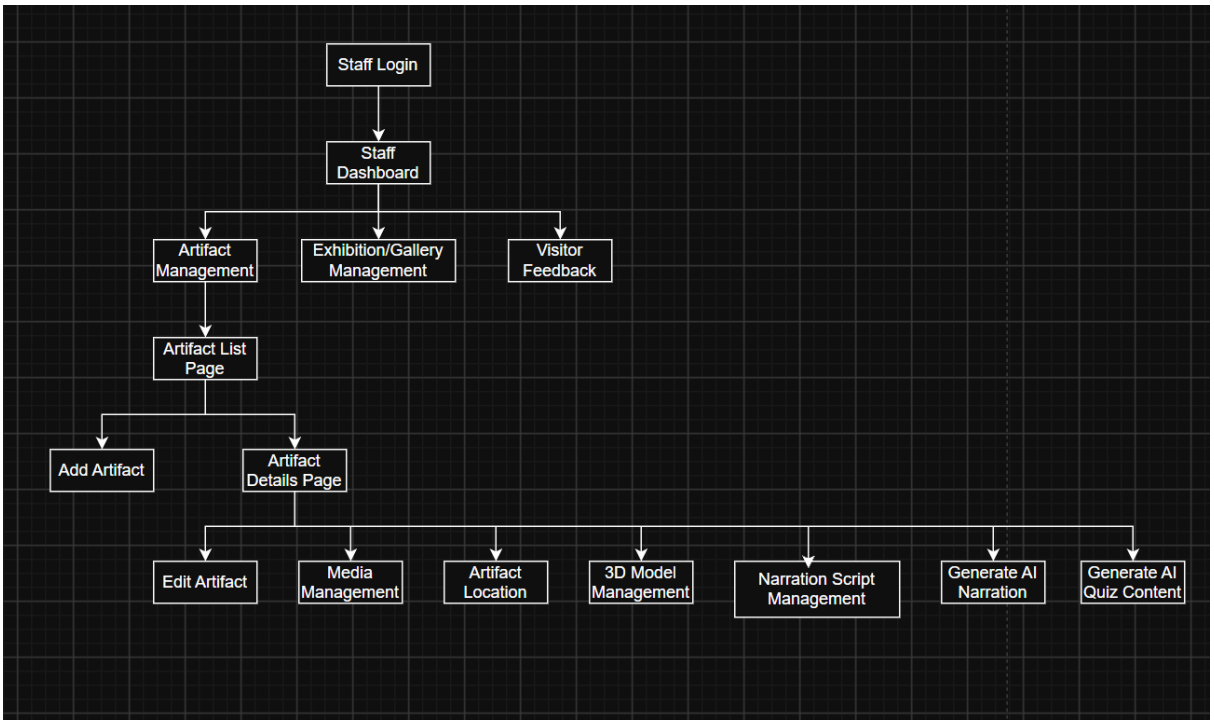
The System Functional Overview presents the main user-facing screens, navigation relationships, role-based screen access, background system functions, and data entities of the Smart Museum Guide System (SMGS). The system consists of a Flutter mobile application for Visitors and one shared React web portal for four operational roles: Administrators, Museum Staff, Curators, and System Staff. It also integrates with external AI services, payment gateways, Google Maps services, and supporting backend services.

3.1.1 Screens Flow

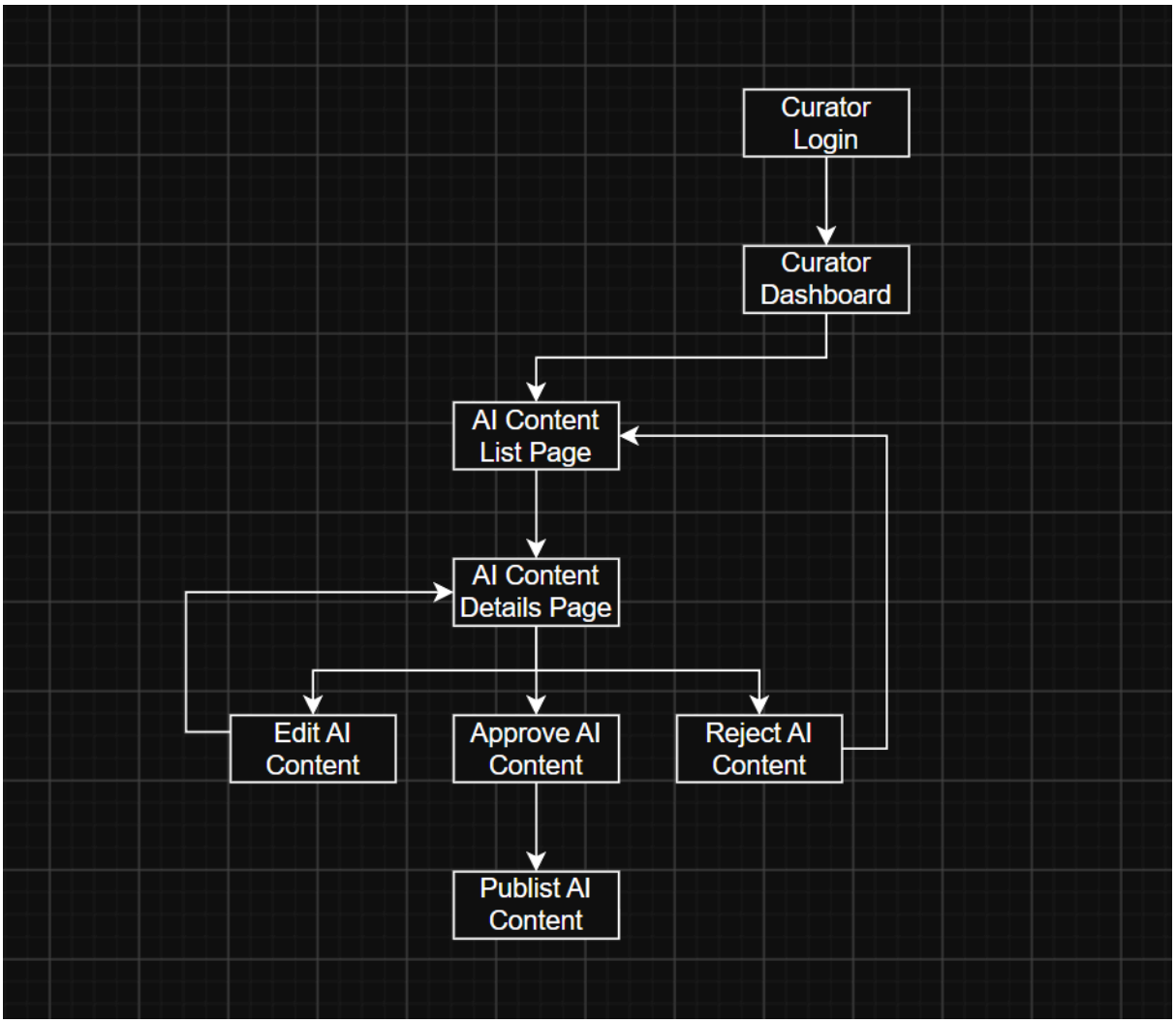
A. Visitor Mobile App


User Login
Home Page
Museum List Page
Artifact Search
Tour Preference
Visitor History
Language Settings
Visitor Rating & Feedback
Quiz History
Museum Details Page
Search Results Page
Personalized TourRecommendation
3D Indoor Museum Navigation
Exhibition/Gallery List
Artifact List Page
Google Maps Museum Directions
QR Code Scanner
Image Recognition
Map/Artifact List
Artifact Details Page
3D Artifact Viewer
AI Narration
AI Chat/Q&A
Quiz
Quiz Result
Ticket Services
Purchase Ticket
Payment Result
Link/Verify Ticket
Ticket Verification Result
Premium Services
Service Details
Checkout
Payment Result
Premium Content Access

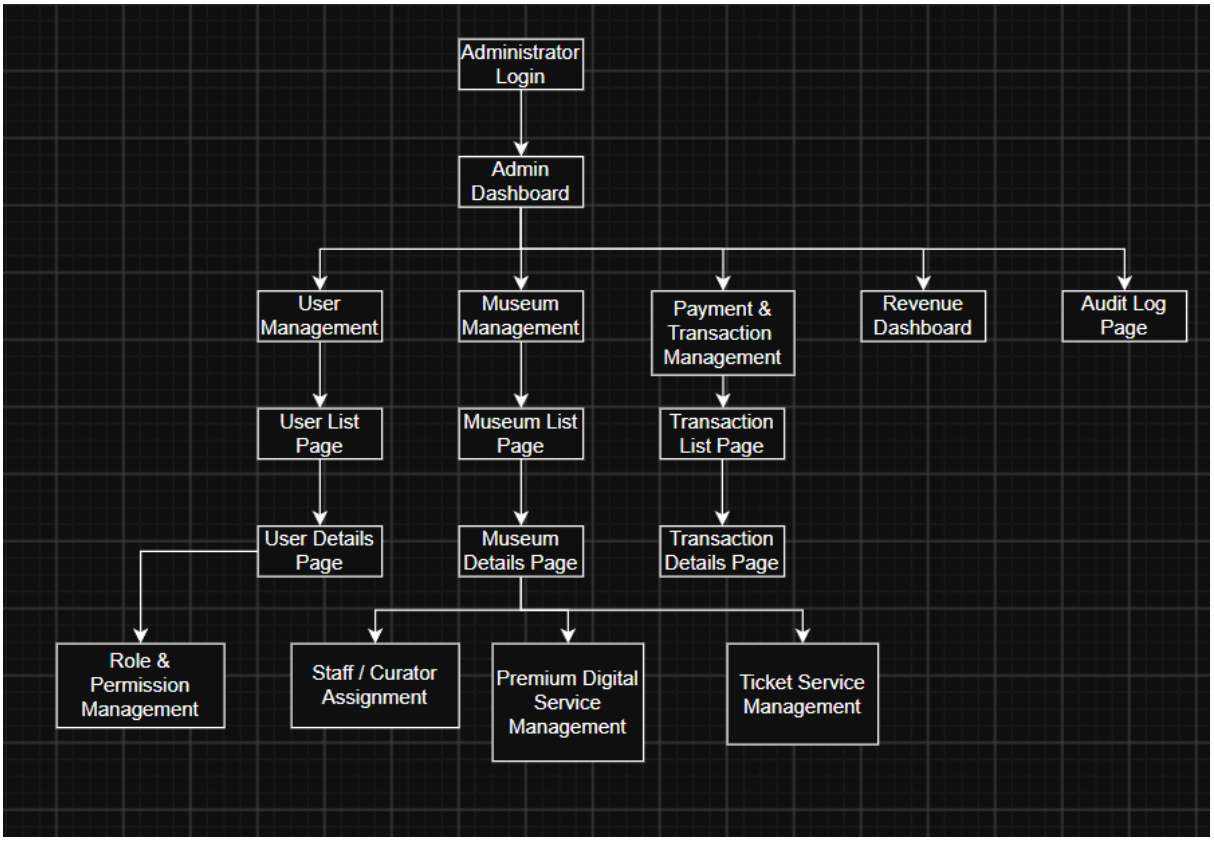
B. Museum Staff Web



C. Staff Web


Curator Login
Curator Dashboard
AI Content List Page
AI Content Details Page
Edit AI Content
Approve AI Content
Reject AI Content
Publist AI Content

D. Platform Administrator Web



3.1.2 Screen Descriptions

A. Visitor Mobile Application

#	Feature	Screen	Description
1	FE-12 Authentication & Access Control	User Login	Allows visitors to enter account credentials and access the SMGS mobile application.
2	General	**Home Page**	Provides the main navigation entry to museum browsing, artifact search, personalized tours, history, feedback, quiz history, and language settings.
3	FE-09 Search & Browse	Museum List Page	Displays the museums currently supported by SMGS and allows the visitor to select a museum.
4	FE-09 Search & Browse	Museum Details Page	Displays information about the selected museum and provides access to exhibitions, artifacts, directions, ticket services, and premium digital services.
5	FE-09 Search & Browse	Artifact Search	Allows visitors to enter keywords and search for supported museum artifacts.
6	FE-09 Search & Browse	Search Results Page	Displays artifacts matching the visitor's search criteria and allows an artifact to be selected.

7	FE-05 Tour & Navigation	Tour Preference Setup	Allows visitors to provide interests and available visiting time for personalized tour recommendation.
8	FE-05 Tour & Navigation	Personalized Tour Recommendation	Displays a recommended museum tour and highlighted artifacts based on the visitor's interests and available time.
9	FE-05 Tour & Navigation	3D Indoor Museum Navigation	Displays the supported indoor museum map and route information for navigating between tour locations.
10	FE-10 Visitor History & Feedback	Visitor History	Displays artifacts previously visited or explored by the authenticated visitor.
11	FE-06 Multilingual Support	Language Settings	Allows the visitor to select a supported interface and content language, initially Vietnamese or English.
12	FE-10 Visitor History & Feedback	Visitor Rating & Feedback	Allows visitors to submit ratings and comments about artifacts or their museum experience.
13	FE-10 Visitor History & Feedback	Quiz History	Displays quiz results previously recorded for the authenticated visitor.
14	FE-09 Search & Browse	Exhibition / Gallery List	Displays exhibitions or galleries belonging to the selected museum for further browsing.
15	FE-09 Search & Browse	Artifact List Page	Displays artifacts associated with the selected exhibition or gallery
16	FE-05 Tour & Navigation	Google Maps Museum Directions	Displays or opens Google Maps directions to the selected supported museum.
17	FE-01 Artifact Discovery	QR Code Scanner	Uses the mobile device camera to scan a QR code and identify the corresponding artifact.
18	FE-01 Artifact Discovery	Artifact Image Recognition	Allows the visitor to capture or provide an artifact image for recognition within the supported artifact dataset.
19	FE-01 Artifact Discovery	Museum Map / Artifact List	Allows visitors to locate and select an artifact manually from the supported museum map or artifact list.
20	FE-01 Artifact Discovery	Artifact Details Page	Displays detailed information about the selected artifact and provides access to 3D visualization, AI narration, and AI Q&A. It may also provide an optional quiz entry point, while visitors can access quizzes later from the app's quiz section rather than using the phone continuously during the visit.
21	FE-01 Artifact Discovery	3D Artifact Viewer	Displays the interactive 3D model of a supported artifact when a 3D model is available.
22	FE-02 AI Narration	AI Narration	Presents approved AI-assisted narration and explanations for the selected artifact.
23	FE-02 AI Narration	AI Chat / Q&A	Allows visitors to ask follow-up questions and receive conversational explanations related to the selected artifact.
24	FE-03 Quiz & Feedback	Artifact Quiz	Displays optional interactive quiz questions related to the selected artifact and allows the visitor to answer and

			submit them at their own pace, including after the museum visit.
25	FE-03 Quiz & Feedback	Quiz Result	Displays the visitor's quiz score and instant feedback after optional quiz completion, and saves the result for later review.
26	FE-11 Ticket & Payment	Ticket Services	Provides access to museum ticket purchasing and existing-ticket linking or verification functions.
27	FE-11 Ticket & Payment	Museum Ticket Purchase	Allows the visitor to select and purchase a supported museum ticket through the configured payment process.
28	FE-11 Ticket & Payment	**Existing Ticket Verification**	Allows the visitor to link or verify an existing museum ticket supported by SMGS.
29	FE-11 Ticket & Payment	Ticket Verification Result	Displays whether an existing museum ticket was successfully verified or could not be validated.
30	FE-11 Ticket & Payment	Payment Result	Displays the result of a ticket or premium-service payment transaction.
31	FE-11 Ticket & Payment	Premium Digital Services	Displays premium digital services available for the selected museum.
32	FE-11 Ticket & Payment	Premium Service Details	Displays the description, access benefits, validity information, and price of a selected premium digital service.
33	FE-11 Ticket & Payment	Premium Service Checkout	Displays purchase information and allows the visitor to confirm payment for the selected premium service.
34	FE-11 Ticket & Payment	Premium Content Access	Provides access to premium digital content or services when the visitor has a valid entitlement for the selected day.

B. Museum Staff Web Application

#	Feature	Screen	Description
1	FE-12 Authentication & Access Control	Museum Staff Login	Allows an authorized museum staff member to sign in to the SMGS museum staff management interface.
2	FE-07 Museum Content Management	Museum Staff Dashboard	Provides the main navigation entry to artifact management, artifact translation management, ticket type management, and ticket verification functions for the assigned museum.

3	FE-07 Museum Content Management	Artifact Management	Provides access to basic artifact records belonging to the museum assigned to the staff member.
4	FE-07 Museum Content Management	Artifact List Page	Displays the list of artifacts managed by the assigned museum and allows staff to select an artifact or create a new basic artifact record.
5	FE-07 Museum Content Management	Add Artifact	Allows staff to enter and save basic information for a new museum artifact.
6	FE-07 Museum Content Management	Edit Basic Artifact Record	Allows authorized museum staff to update basic information of an existing artifact record.
7	FE-06 Multilingual Support	Artifact Translation Management	Allows museum staff to manage basic artifact content translations for supported languages.
8	FE-11 Ticket & Payment	Ticket Type Management	Allows museum staff to create, edit, and configure ticket types and pricing settings.
9	FE-11 Ticket & Payment	Existing Ticket Verification / Check-in	Allows museum staff to verify visitor tickets and record museum check-in.

C. Curator Web Application

#	Feature	Screen	Description
1	FE-12 Authentication & Access Control	Curator Login	Allows an authorized Curator to sign in to the web portal.
2	FE-08 AI Content & Review	Curator Dashboard	Provides the main entry for content review queues, approval histories, and publishing controls within assigned museum scope.

3	FE-08 AI Content & Review	Content Review Queue	Displays submitted artifact content drafts, narration scripts, and quiz questions awaiting Curator review.
4	FE-08 AI Content & Review	Content Review Details	Allows the Curator to inspect full details, record review notes, reject content, or approve draft content for publication.
5	FE-08 AI Content & Review	Publish Content	Allows the Curator to publish approved content to make it officially visible to Visitors in the mobile app.

E. Platform Administrator Web Application

#	Feature	Screen	Description
1	FE-12 Authentication & Access Control	Administrator Login	Allows a platform administrator to sign in to the SMGS administration interface.
2	FE-12 Authentication & Access Control	Admin Dashboard	Provides the main navigation entry to user, museum, transaction, revenue, and audit-management functions.
3	FE-12 Authentication & Access Control	User Management	Provides access to system user accounts and user-related administration functions.
4	FE-12 Authentication & Access Control	User List Page	Displays registered system users and allows the administrator to select a user for further management.
5	FE-12 Authentication & Access Control	User Details Page	Displays detailed information about a selected user account and provides access to role and permission configuration.
6	FE-12 Authentication & Access Control	Assign User Role & Permission	Allows the administrator to assign or update roles and permissions for a selected user.

7	FE-12 Authenticatio n & Access Control	Museum Management	Provides access to supported museum records and museum-level administration functions.
8	FE-12 Authenticatio n & Access Control	Museum List Page	Displays museums currently configured in the SMGS platform.
9	FE-12 Authenticatio n & Access Control	Museum Details Page	Displays detailed information about a selected museum and provides access to museum-related administrative functions.
10	FE-12 Authenticatio n & Access Control	Staff Assignment	Allows the administrator to assign Museum Staff or Curators to a supported museum.
11	FE-11 Ticket & Payment	Premium Digital Service Management	Allows the administrator to manage premium digital services configured for a supported museum.
12	FE-11 Ticket & Payment	Ticket Service Management	Allows the administrator to manage ticket-related services configured for a supported museum.
13	FE-11 Ticket & Payment	Payment & Transaction Management	Provides access to purchase, payment, and transaction records managed by the platform.
14	FE-11 Ticket & Payment	Transaction List Page	Displays payment and transaction records for administrative monitoring and management.
15	FE-11 Ticket & Payment	Transaction Details Page	Displays detailed information about a selected transaction, including its related purchase/payment information and transaction status.
16	FE-11 Ticket & Payment	Revenue Dashboard	Displays revenue-related information derived from premium services, ticket services, purchases, and transactions.
17	FE-12 Authenticatio n & Access Control	Audit Log Page	Displays recorded system and administrative activities for monitoring and auditing purposes.

3.1.3 Screen Authorization

A. Visitor Mobile Application

Screen	Visitor	Museum Staff	Curator	System Staff	Administrator
User Login	X				
Home Page	X				
Museum List Page	X				
Museum Details Page	X				
Artifact Search	X				
Search Results Page	X				
Exhibition / Gallery List	X				
Artifact List Page	X				
QR Code Scanner	X				
Artifact Image Recognition	X				
Museum Map / Artifact List	X				
Artifact Details Page	X				
3D Artifact Viewer	X				
AI Narration	X				
AI Chat / Q&A	X				
Artifact Quiz	X				
Quiz Result	X				
Tour Preference Setup	X				
Personalized Tour Recommendation	X				
3D Indoor Museum Navigation	X				
Google Maps Museum Directions	X				
Visitor History	X				
Quiz History	X				
Visitor Rating & Feedback	X				
Language Settings	X				
Ticket Services	X				
Museum Ticket Purchase	X				
Existing Ticket Verification	X				
Ticket Verification Result	X				
Payment Result	X				
Premium Digital Services	X				
Premium Service Details	X				
Premium Service Checkout	X				
Premium Content Access	X				

B. Museum Staff Web Application

Screen	Visitor	Museum Staff	Curator	System Staff	Administrator
Museum Staff Login		X			
Museum Staff Dashboard		X			
Artifact Management		X			
Artifact List Page		X			
Add Artifact		X			
Edit Basic Artifact Record		X			
Artifact Translation Management		X			
Ticket Type Management		X			
Existing Ticket Verification / Check-in		X			

C. Curator Web Application

Screen	Visitor	Museum Staff	Curator	System Staff	Administrator
Curator Login			X		
Curator Dashboard			X		
Content Review Queue			X		
Content Review Details			X		
Publish Content			X		

D. System Staff Web Application

Screen	Visitor	Museum Staff	Curator	System Staff	Administrator
System Staff Login				X	
System Support Dashboard				X	
Feedback & Complaint Moderation				X	
Generate AI Narration				X	
Generate AI Quiz Content				X	
Transaction Reconciliation & Refunds				X	

D. System Staff Web Application

#	Feature	Screen	Description
1	FE-12 Authentication & Access Control	System Staff Login	Allows System Staff to sign in to the web portal.
2	System Support & Moderation	System Support Dashboard	Provides central access to visitor support requests, complaints, refund cases, feedback moderation, and operational incident logs.
3	FE-10 Visitor History & Feedback	Feedback & Complaint Moderation	Allows System Staff to inspect visitor ratings, moderate inappropriate feedback, and resolve visitor complaints.
4	FE-11 Ticket & Payment	Transaction Reconciliation & Refunds	Allows System Staff to handle payment dispute cases, process refund requests, and reconcile transaction anomalies with payment gateways.

Screen	Visitor	Museum Staff	Curator	System Staff	Administrator
Administrator Login					X
Admin Dashboard					X
User Management					X
User List Page					X
User Details Page					X
Assign User Role & Permission					X
Museum Management					X
Museum List Page					X
Museum Details Page					X
Staff Assignment					X
Premium Digital Service Management					X
Ticket Service Management					X
Payment & Transaction Management					X
Transaction List Page					X
Transaction Details Page					X
Revenue Dashboard					X
Audit Log Page					X

3.1.4 Non-Screen Functions

#	Feature	System Function	Description
1	FE-01 Artifact Discovery	Artifact Image Recognition Service	Processes an image captured or uploaded by a visitor and identifies the corresponding artifact from the supported artifact dataset.
2	FE-02 AI Narration & Q&A	AI Question Answering Service	Processes visitor questions and generates contextual answers based on published information related to the selected artifact.
3	FE-03 Quiz	Quiz Evaluation Service	Evaluates submitted quiz answers, calculates the result, and returns instant feedback to the visitor. The service supports optional self-paced quizzes and does not require quiz use during the physical visit.
4	FE-04 Gamification	Gamification Processing Service	Calculates and updates visitor points, badges, and achievements based on supported activities as supplementary engagement features. Points and badges are not redeemable for external rewards within the current scope.
5	FE-05 Tour & Navigation	Personalized Tour Recommendation Service	Generates a suitable tour route and highlighted artifacts based on visitor interests and available visiting time.
6	FE-05 Tour & Navigation	Indoor Route Calculation Service	Calculates supported navigation routes between museum locations using the configured indoor museum map.
7	FE-05 Tour & Navigation	Google Maps Integration Service	Integrates with Google Maps to provide outdoor navigation directions to a supported museum venue.

8	FE-08 AI Content & Review	AI Content Generation Service	Generates draft AI-assisted narration and quiz content for artifacts before staff review and approval.
9	FE-11 Ticket & Payment	Payment Gateway Integration Service	Sends payment requests to the configured online payment gateway and receives payment transaction results.
10	FE-11 Ticket & Payment	Ticket Verification Service	Validates or links supported museum ticket information and returns the ticket verification result.
11	FE-11 Ticket & Payment	Premium Access Entitlement Service	Grants and validates access to purchased premium digital services for the selected valid day.
12	FE-12 Authentication & Access Control	Authentication & Authorization Service	Verifies user identity and determines access permissions according to the user's assigned role.
13	FE-12 Authentication & Access Control	Single Active Session Control	Ensures that only one active device session is allowed for each user account at a time.
14	FE-12 Authentication & Access Control	Museum Assignment Authorization	Restricts Museum Staff and Curator operations to the museum scope to which they are assigned.
15	FE-12 Administration	Audit Logging Service	Records important administrative and system activities for later monitoring and auditing.
16	System Support	**Content Cache Service**	Caches selected key artifact content to provide limited access when network connectivity is unstable.

3.1.5 Entity Relationship Diagram



Link: https://dbdiagram.io/d/SMGS-6aaf32cc6d586d617e766aae

Database Entities Summary (smgs_schema.dbml):

#	Entity	Description
1	Stores user credentials and profile details for Visitors, Museum Staff, Curators, System Staff, and Administrators.	
2	Role	Defines system role titles (Visitor, MuseumStaff, Curator, SystemStaff, Administrator).
	Permission	
	UserRole	Associates users with their assigned roles. In the current release, each user account is assigned one primary role.
6	AuditLog	Records important user and system activities for security, monitoring, and auditing purposes.
7	Gallery	Stores information about museum galleries where artifacts are displayed or organized.
8	Theme	Defines thematic categories used to organize and browse museum artifacts.
9	Artifact	Stores core information about museum artifacts, including title, description, location, year, and status.
10	ArtifactTheme	Associates artifacts with one or more themes.
11	Media	Stores media resources associated with artifacts, such as images, videos, and other multimedia content.
12	Language	Defines languages supported by the system for multilingual user interfaces and museum content.
14	Narration	Stores AI-generated or staff-managed narration scripts and audio content for artifacts in different languages.
17	Quiz	Stores quizzes associated with museum artifacts and their basic information.
18	QuizQuestion	Stores individual questions belonging to a quiz.
19	QuizOption	Stores answer options for quiz questions and identifies the correct option.
20	QuizAttempt	Records a visitor's attempt to complete a quiz, including score and completion time.
21	QuizAnswer	Stores the answer selected by a visitor for each question in a quiz attempt.
23	Feedback	Stores visitor ratings and comments about artifacts or their museum experience.
25	Badge	Defines badges and achievements that visitors can unlock through specified activities or requirements as supplementary, non-redeemable in-app recognition within the current scope.
26	UserBadge	Records badges earned by individual visitors and the time they were obtained.

#	Entity	Description
27	Tour	Stores information about museum tours, including tour name, description, duration, and premium status.
	TourStop	
29	UserInterest	Stores visitor interests used to personalize artifact and tour recommendations.
30	TourRecommendation	Records personalized tour recommendations generated for individual visitors.
33	Invoice	Stores billing information associated with a visitor's purchase of premium content or tours.
		35 ArtifactContentVersion Tracks versioning, draft states, and status transitions (draft, pending_review, published, rejected, archived) for artifact content. 36 NarrationVersion Stores historical versions and draft revisions of AI-generated or staff-crafted narration scripts. 37 PaymentTransaction Records detailed payment transaction logs and gateway reconciliation statuses. 38 AccessRight Defines digital service and ticket entitlement access rights granted to visitors after verified payment completion. 39 Refund Stores refund requests, approval statuses, and gateway settlement records managed by System Staff. 40 Complaint Tracks visitor complaints, support tickets, and resolution history handled by System Staff.

3.2 Functional Requirements List

3.2.1 FR-01: Discover Artifact by QR Code or Image Recognition

The system allows Visitors on the mobile application to scan a physical QR code or upload an artifact image to identify the corresponding artifact record and open its details.

3.2.2 FR-02: View Published Artifact Content

The system displays artifact information, narration scripts, 3D visualization, and quizzes only when the content status is published. Content in draft, pending_review, rejected, or archived states is not accessible to Visitors.

3.2.3 FR-03: Create and Submit Content Draft

Museum Staff can create new artifact details, narration scripts, and quiz questions, save them with status draft, and submit them for review, changing the status to pending_review within assigned museum scope.

3.2.4 FR-04: Review and Publish/Reject Content

Curators inspect content drafts in pending_review state. Curators can reject drafts with notes (changing status to rejected) or approve and publish them (changing status to published).

3.2.5 FR-05: Purchase Product and Grant Access Right

Visitors can purchase tickets or premium digital guide services via the integrated payment gateway. Upon successful payment verification, the system creates an AccessRight record granting digital service access.

3.2.6 FR-06: Submit Support or Refund Request

Visitors can submit support tickets, complaints, or refund applications for purchased services directly through the mobile application.

3.2.7 FR-07: Process Refund and Reconcile Transaction

System Staff handle refund applications, execute refunds through the payment gateway interface, and reconcile local PaymentTransaction entries with gateway financial summaries.

3.2.8 FR-08: Get Google Maps Directions

Visitors can request outdoor navigation to supported museums that have the googleMapsVisible flag enabled, launching Google Maps directions to the venue.

4. Non-Functional Requirements

4.1 External Interfaces

- The system shall provide a mobile application for visitors.
- Museum Staff, Curators, System Staff, and Administrators shall access the system through web-based interfaces.
- The mobile application shall use the device camera for QR-code scanning and artifact image recognition.
- The system shall integrate with AI services, Google Maps, and the configured payment gateway through APIs.
- Communication between the frontend, backend, and external services shall use secure HTTPS connections.
- The backend shall communicate with the PostgreSQL database for storing and retrieving system data.

4.2 Quality Attributes

4.2.1 Usability

- The mobile application shall provide an intuitive interface that requires minimal guidance for visitors.
- Navigation, buttons, labels, and screen layouts shall remain consistent throughout the application.
- The system shall provide Vietnamese user interfaces by default for all visitor and staff applications in the current release. Multilingual public interface switching is reserved for future scope.
- Validation and error messages shall clearly indicate problems and required corrective actions.

4.2.2 Reliability

- The system should achieve at least 99.5% availability during museum operational hours, excluding planned maintenance and failures of external services.
- The system shall maintain data consistency during payment processing, content updates, and other critical operations.
- Failed or cancelled payments shall not create active premium-service access.
- Failures of AI services, Google Maps, or payment services shall not corrupt existing system data.

4.2.3 Performance

- Standard API requests such as artifact search, artifact information retrieval, and content loading should normally complete within 2 seconds under standard network conditions.
- QR-code identification should return artifact information within 3 seconds under normal network conditions.
- Artifact image recognition should normally return a result within 10 seconds.

- AI responses may require additional processing time depending on the external AI service.
- The system shall support multiple concurrent visitors without significant service degradation.
- Images, media, and 3D models should be optimized to reduce loading time and device resource usage.

4.2.4 Security

- Users shall be authenticated before accessing protected functions.
- The system shall apply role-based access control for Visitors, Museum Staff, Curators, System Staff, and Administrators.
- Museum Staff and Curators shall only access data belonging to their assigned museum scope.
- Passwords shall not be stored in plaintext.
- Protected communication shall use HTTPS.
- Passwords, API keys, access tokens, and sensitive payment information shall not be stored in audit logs.

4.2.5 Maintainability

- Source code shall follow consistent coding conventions and naming standards.
- The system shall separate frontend, backend, database, AI, and external-service components.
- Source code shall be managed using Git and GitHub.
- Sensitive configuration and credentials shall be separated from source code.
- Important APIs and modules shall be documented to support maintenance and future development.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Only published museum content is visible to visitors in the mobile application.
BR-02	AI-generated content drafts must be reviewed and published before being accessible to visitors.
BR-03	Curators review and publish museum content; Museum Staff create drafts within assigned museum scope.
BR-04	Free museum content is available to visitors without payment; premium services require a valid entitlement.
BR-05	Failed or cancelled payments shall not grant active access to premium services or tickets.
BR-06	Each completed transaction must have a unique reference code from the payment gateway.
BR-07	The system must prevent duplicate processing of payment notification callbacks.
BR-08	Visitors can only view their own visit history, quiz results, and purchases.

BR-09	Visitor ratings must be integer values between 1 and 5 stars.
BR-10	Audit logs must not contain plaintext passwords or sensitive financial credentials.
BR-11	System Staff handle support tickets, complaints, refund cases, and feedback moderation.
BR-12	Administrators manage accounts, roles, museum assignments, public Google Maps visibility flags (googleMapsVisible), and system-wide configurations.

5.2 Common Requirements

All dates and timestamps should use one consistent application timezone convention.

All user-facing validation messages should identify the field or action that requires correction.

All protected APIs must perform server-side authorization even when the UI hides unauthorized controls.

Content Status must follow the exact lifecycle: draft, pending_review, published, rejected, and archived.

External-service failures should not corrupt local transaction or content state.

5.4 Integration & Implementation Status

User authentication, museum discovery, outdoor route retrieval (via Google Maps integration), and selected artifact/content management APIs are integrated with backend services and PostgreSQL persistence. Several Visitor interactive flows, payment gateway checkout demos, refund processing steps, quiz evaluations, and operational dashboard analytics currently utilize mock/prototype data for demonstration purposes.

5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG01	In line	No artifact is found when searching	*Không tìm thấy hiện vật nào.*
2	MSG02	In red, under the text box	A required field is empty	*Trường này là bắt buộc.*
3	MSG03	Toast message	User login is successful	*Đăng nhập thành công.*
4	MSG04	In line	Email or password is incorrect	*Email hoặc mật khẩu không chính xác. Vui lòng thử lại.*
5	MSG05	Toast message	QR code is scanned successfully	*Nhận diện hiện vật thành công.*
6	MSG06	Toast message	QR code cannot be recognized	*Không thể nhận diện hiện vật từ mã QR này.*

7	MSG07	Toast message	Image recognition succeeds	*Nhận diện hiện vật qua hình ảnh thành công.*
8	MSG08	Toast message	Image recognition fails	*Không thể nhận diện hiện vật. Vui lòng thử ảnh khác.*
9	MSG09	In line	AI Guide cannot generate a response	*Trợ lý AI tạm thời không khả dụng. Vui lòng thử lại sau.*
10	MSG10	Toast message	Quiz is submitted successfully	*Nộp bài trắc nghiệm thành công.*
11	MSG11	In line	No quiz is available for the selected artifact	*Chưa có bài trắc nghiệm cho hiện vật này.*
12	MSG12	Toast message	User earns points after completing an activity	*Tích điểm thành công.*
13	MSG13	Toast message	User earns a new badge	*Xin chúc mừng! Bạn đã đạt được huy hiệu mới.*
14	MSG14	In red, under the text box	Available tour time is invalid	*Vui lòng nhập thời gian tham quan hợp lệ.*
15	MSG15	Toast message	Personalized tour is generated successfully	*Tạo lộ trình tham quan cá nhân hóa thành công.*
16	MSG16	In line	Personalized tour cannot be generated	*Không thể tạo lộ trình tham quan với các tuỳ chọn đã chọn.*
17	MSG17	Toast message	Visit session starts successfully	*Bắt đầu chuyến tham quan thành công.*
18	MSG18	In line	Current indoor location cannot be determined	*Vị trí hiện tại của bạn không khả dụng.*
19	MSG19	Toast message	Purchase order is created successfully	*Tạo đơn hàng thành công.*
20	MSG20	Toast message	Payment is successful	*Thanh toán hoàn tất thành công.*

21	MSG21	Toast message	Payment fails	*Thanh toán thất bại. Vui lòng thử lại.*
22	MSG22	In line	Payment is still being processed	*Giao dịch thanh toán của bạn đang được xử lý.*
23	MSG23	Toast message	Digital guide access is activated	*Kích hoạt quyền truy cập thuyết minh số thành công.*
24	MSG24	Toast message	Museum ticket is issued successfully	*Phát hành vé tham quan thành công.*
25	MSG25	Toast message	Content is saved as a draft	*Lưu bản nháp thành công.*
26	MSG26	Toast message	Content is submitted for review	*Gửi yêu cầu duyệt nội dung thành công.*
27	MSG27	Toast message	Content is approved	*Phê duyệt nội dung thành công.*
28	MSG28	Toast message	Content is rejected	*Nội dung đã bị từ chối. Vui lòng kiểm tra góp ý và cập nhật lại.*
29	MSG29	Toast message	Approved content is published	*Xuất bản nội dung thành công.*
30	MSG30	Toast message	Artifact information is created successfully	*Tạo mới thông tin hiện vật thành công.*
31	MSG31	Toast message	Artifact information is updated successfully	*Cập nhật thông tin hiện vật thành công.*
32	MSG32	Toast message	User submits feedback successfully	*Cảm ơn bạn đã gửi phản hồi.*
33	MSG33	In red, under the text box	Input exceeds the maximum allowed length	*Độ dài tối đa cho phép là {max_length} ký tự.*
34	MSG34	Confirmation dialog	User attempts to delete or discard unsaved content	*Bạn có chắc chắn muốn hủy bỏ các thay đổi chưa lưu này không?*

35	MSG35	Confirmation dialog	Staff attempts to publish approved content	*Bạn có chắc chắn muốn xuất bản nội dung này không?*

5.5 Other Requirements

5.5.1 Role Mapping Appendix

The following table maps organizational labels, database roles, backend JWT roles, and React frontend internal roles across the platform:

Business Label	Database Role	Backend JWT Role	React Internal Role
Visitor	Visitor	Visitor	visitor
Museum Staff	MuseumStaff	Staff	museumStaff
Curator	Curator	Staff	curator
System Staff	SystemStaff	Staff	systemStaff
Administrator	Administrator	Admin	admin